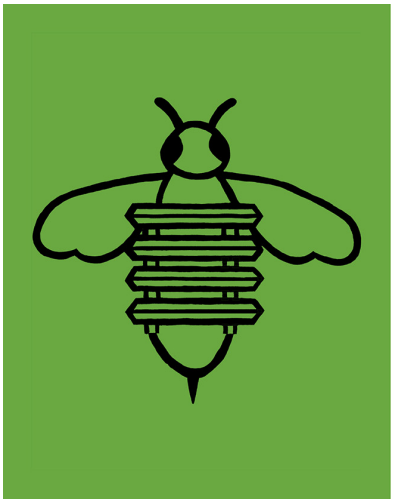


ALL BODILY ACTIVITY DEPENDS ON TEMPERATURE

THERMOREGULATION IN INSECTS

IN CONTEXT

KEY FIGURE

Bernd Heinrich (1940–)

BEFORE

1837 In the UK, George Newport observes that flying insects are capable of raising their body temperature above the ambient temperature.

1941 Danish researchers August Krogh and Eric Zeuthen conclude that the temperature of an insect's flight muscles just before takeoff determine the muscles' rate of work during flight.

AFTER

1991 German biologist Harald Esch describes how muscle "warm-up" plays a role in brood incubation and colony defense as well as flight preparation.

2012 Using infrared thermography, Spanish zoologist José R. Verdú shows how some dung beetle species heat or cool their thorax to improve flight performance.

Insects are usually described as "cold-blooded," or ectotherms. Unlike mammals and other "warm-blooded" endotherms, animals that maintain